DES643 Project

Team Virtual

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Final Logbook

Project: Immersive Space Exploration System

Week 1: Problem Understanding & Ideation

Work Done

- Identified the core problem in astronomy education:
    - Difficulty in understanding scale, distance, and spatial relationships
- Studied how VR can improve learning through immersion
- Reviewed concepts of:
    - Presence
    - Perception
    - Cognitive load

Key Decisions

- Chose to build a VR-based space exploration system
- Decided to create a custom binary star system instead of the solar system

Outcome

Finalized project idea and learning objectives
Defined user goal: interactive exploration instead of passive learning

Week 2: System Design & Concept Development

Work Done

- Designed the overall system architecture
- Created conceptual layout:
    - Binary stars (Solari & Lumina)
    - 5 planets with different environments
- Developed initial storyboard and user journey

Design Focus

- Learner-centered approach
- Exploration-based interaction
- Reducing cognitive load

Concepts Applied

- Spatial design principles (scale, depth, navigation)
- Design thinking approach (Define–Ideate)

Outcome

Clear structure of VR world and interaction flow ready

Week 3: Prototyping & Unity Setup

Work Done

- Set up project in **Unity 3D**
- Learned and implemented:
    - Scenes and GameObjects
    - Prefabs
    - Lighting and camera systems
- Imported initial assets

Technical Work

- Started building base environment
- Configured VR setup using XR tools

Concepts Applied

- Unity workflow (Scenes, Components, Prefabs)

Outcome

Basic VR environment with navigation ready

Week 4: Environment & Planet Development

Work Done

- Designed and implemented planetary environments:
  - Pyronis (barren)
  - Sulvara (toxic atmosphere)
  - Dravix (rocky desert)
  - Zephyrus (gas giant)
  - Thalor (ocean storm world)

Techniques Used

- Fog and atmospheric effects
- Texturing and materials
- Particle systems (dust, rain, storms)

Design Focus

- Visual diversity
- Environmental storytelling
- Realistic but optimized rendering

Outcome

Fully developed immersive planetary environments

Week 5: Core System Implementation

Work Done

- Implemented **binary star system**:
  - Solari (yellow star)
  - Lumina (white star)
- Developed orbital motion logic
- Added lighting interactions (dual shadows)

Interaction Features

- Free navigation (fly-through system)
- Scale transitions (planet ↔ system view)

Concepts Applied

- Perception & presence in VR
- Spatial immersion and user orientation

Outcome

Functional VR system with dynamic celestial interactions

Week 6: Interaction, Optimization & Finalization

Work Done

- Added:
  - Object interaction (info panels)
  - UI system (selection menu)
- Optimized performance:
  - Reduced heavy rendering
  - Used fog, shaders, layered textures

Challenges Solved

- Transparency and depth issues
- Performance constraints in VR

- Maintaining immersion without lag

Final Outcome

Complete VR-based space exploration system ready

Overall Learnings

Technical

- Unity VR development
- Environment design
- Optimization techniques

Design

- Importance of human-centered design
- Spatial design is critical in VR
- Iteration improves experience quality

Conceptual

- VR reduces cognitive load in learning
- Immersion improves understanding of complex systems

Conclusion

The project successfully demonstrates how virtual reality can transform abstract concepts into immersive experiences.

By enabling users to explore a custom binary star system, the VRSES enhances spatial understanding, engagement, and learning effectiveness—highlighting the strong potential of VR in education.